

Physics-Informed Joint State Estimation, Event Detection, and Source Localization From Sparse PMUs Under Limited Observability

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ABSTRACT Sparse phasor measurement unit (PMU) coverage couples three questions that are often evaluated separately: what is happening in the unobserved network, which event occurred, and where it originated. We formulate these questions as joint state, physical-intervention, and measurement-integrity inference in a hybrid differential-algebraic model. The formulation separates physical source hypotheses from missing or corrupted measurements and introduces two target-specific limits: functional observability of hidden voltages and projected pairwise information for event-source diagnosability. Evidence comes from two frozen IEEE 39-bus campaigns using the same eight PMU buses but different simulators and contracts. On 100 PowerDynamics nonlinear test trajectories, a causal Kalman estimator reduces mean hidden-bus total vector error from 0.3402% to 0.009634%; a preregistered functional residual correlates with bus-level error ($\rho = 0.5625$). A measurement-only static physical MAP closes 97.4%–98.3% of the nominal-center error gap on 60 operating-point-shifted test plants despite nuisance-information rank 25/43. On a separate 690-trajectory ANDES event bank, a physics-guided hierarchical baseline attains detection F1 0.989, event macro-F1 0.905, and physical-source Top-1 0.592 for sources represented during training. Under complete source holdout, however, only 3 of 303 decisions are correct. These results establish the state-reconstruction component and expose source memorization as the binding failure; they do not yet validate the proposed end-to-end joint estimator.

INDEX TERMS dynamic state estimation, event detection, functional observability, limited observability, phasor measurement units, physics-informed inference, source localization.

I. INTRODUCTION

PHASOR measurement units (PMUs) reveal electromechanical transients through synchronized voltage and current phasors, yet only a fraction of the buses and branches in a transmission network are normally instrumented [1], [2]. A disturbance may therefore originate outside the measured region, alter hidden voltages, and propagate to the available devices. At the same time, a missing frame or corrupted channel can resemble a physical transient. A useful diagnosis must answer three distinct questions: what is the network state, which physical or measurement event is present, and where did each applicable event originate?

Strong prior methods solve important subsets of this problem. Dynamic estimators combine physical models with synchrophasors [2], [3]; sparse-PMU classifiers identify events

and faulted assets [4], [5]; and recent learning systems reconstruct hidden trajectories before localization or estimate state and attack location from partial observations [6], [7]. These results make a broad “joint inference” novelty claim untenable. The unresolved contract considered here is narrower: causal inference in an under-observed transmission system when physical interventions and measurement-integrity failures can coexist, the source asset may be absent from source-labeled training, and the sensing geometry may make two candidate sources physically indistinguishable.

This distinction matters because event recognition, state reconstruction, and source localization have different information requirements. A generation change can be detected throughout the network while its source remains ambiguous. A dropped PMU identifies an integrity location directly but

removes information needed to localize a simultaneous physical event. Likewise, a state estimate can have low point error although the nuisance intervention is not uniquely identifiable. Treating all three tasks as one multiclass label hides these differences and encourages scores whose denominator does not match the claimed capability.

We represent a physical event as an intervention hypothesis $h = (c, \ell, \tau, \gamma)$ containing family, source, onset, and magnitude, and represent measurement integrity separately through a time-varying availability operator and an additive corruption process. Each physical candidate produces a counterfactual DAE response. The PMU likelihood ranks those responses after projecting away changes that can be explained by uncertain initial state. This construction yields a pairwise diagnosability quantity that predicts posterior ambiguity and motivates candidate-set output or abstention when the available PMUs cannot separate sources.

The empirical record currently contains two complementary, frozen campaigns. The PowerDynamics campaign audits an IEEE 39-bus hybrid DAE, reconstructs 31 hidden complex bus voltages from eight voltage/current PMUs, and studies model mismatch and physical recentering. The separate ANDES campaign contains 690 trajectories spanning eight abnormal classes plus normal operation and evaluates a physics-guided hierarchical ExtraTrees baseline. The two campaigns share PMU buses and a network family, but not simulator semantics, state coordinates, or split contracts; their metrics are never pooled. The event study is presented as a baseline and failure analysis, not as evidence that the proposed joint posterior has already been executed.

The paper makes four contributions.

- 1) We formulate sparse-PMU monitoring as joint inference over hidden state, physical intervention, physical source, and measurement integrity. The eight event labels are mapped to composable physical and integrity variables rather than treated as eight unrelated phenomena.
- 2) We connect functional observability of the requested voltage field with event-source diagnosability. A finite-horizon projected information measure gives a local theoretical limit: sources whose projected signatures coincide cannot be separated by any decision rule under the stated Gaussian model.
- 3) We consolidate two audited evidence lines. On the state campaign, a preregistered structural residual predicts bus-level reconstruction difficulty and measurement-only physical MAP recovers operating-point-shifted voltages. On the event campaign, multiscale and spatial features improve known-source localization, but complete source holdout collapses to 3/303 exact decisions.
- 4) We define the reproducible comparison required for the next joint experiment: learned-only diagnosis, physics-only hypothesis inference, physics plus learned residual discrepancy, and diagnosability-aware inference with abstention, all evaluated on source-disjoint parents with explicit alarm exposure and PMU loss/join conditions.

The strongest result is also the main limitation. The state component is accurate under the tested nominal and operating-point-shift contracts, and the known-source event baseline attains detection F1 0.989 and event macro-F1 0.905. Yet source localization does not transfer to assets excluded from training. The current manuscript therefore establishes the mathematical and empirical boundary that a joint physics-informed method must cross; it does not convert a proposed end-to-end estimator into a completed result.

The remainder of the paper reviews the closest estimation and localization contracts, defines the hybrid DAE and eight event labels, derives the observability and diagnosability limits, specifies the estimator hierarchy, documents both evidence campaigns, and reports the validated results and unresolved joint-inference gate.

II. RELATED WORK AND EXACT RESEARCH GAP

A. STATE ESTIMATION WITH INCOMPLETE MODELS OR MEASUREMENTS

Power-system dynamic state estimation combines machine, control, load, and network models with high-rate synchronized measurements. The task-force review in [2] separates dynamic, tracking, forecasting-aided, and static formulations and identifies model uncertainty and measurement availability as persistent constraints. Robust Gaussian filters address outliers or non-Gaussian measurement noise [8], [9]; these are different failure modes from a stale network or equilibrium center. Katanic *et al.* estimate differential and algebraic states directly in a partially known nonlinear DAE and derive a topological estimability condition [3]. Nguyen *et al.* instead estimate states and unknown inputs in distributed active networks [10]. Both establish that state and nuisance input can be inferred jointly, but neither uses the event-family/source contract or integrity composition studied here.

Learning methods recover states under sparse or heterogeneous measurements by encoding graph structure, power-flow relations, or a model-based estimator in the representation. Ngo *et al.* combine a graph neural network with physics-based state estimates [11]; Yue *et al.* couple graph learning to a robust distribution-system state estimator and include sparse-PMU line-outage identification [12]. These methods broaden the estimator family, but their distribution-system measurements and targets do not provide a common benchmark for the transmission-event source problem.

B. SPARSE-PMU EVENT AND SOURCE INFERENCE

PMU event identification has been addressed through subspace methods, engineered features, transfer learning, and graph models [13]–[16]. Cheng *et al.* demonstrate label-light online detection on a large field-PMU stream [17], while Abdel Hai *et al.* study transfer when only a small labeled set is available [18]. These works establish strong detection contracts, but event presence is not physical-source recovery. Pandey *et al.* jointly detect, classify, and localize events using synchrophasor measurements and system information [4]. Sparse-PMU methods locate faulted lines, generation

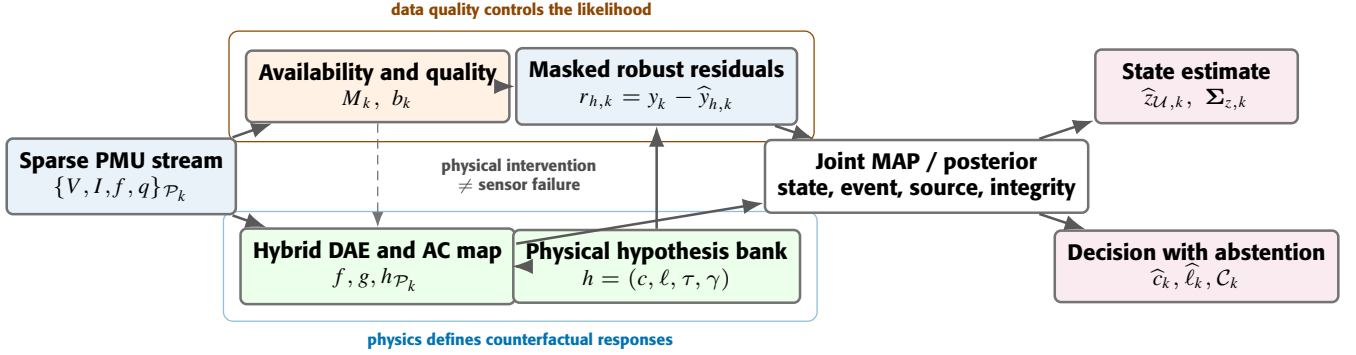


FIGURE 1. Joint inference contract. The physical model generates candidate state and PMU responses, while availability and quality govern which residuals are admissible. Physical interventions and measurement-integrity failures remain separate latent variables. The rightmost outputs distinguish hidden-state uncertainty from event/source ambiguity and permit a candidate set instead of a forced source. Solid arrows describe the proposed online information flow; this paper validates its state-estimation and historical diagnosis components separately.

losses, or outages using learned models and electrical structure [5], [19]–[21]. Group-sparse dictionaries use admittance-, impedance-, and injection-based response models to localize several disturbance types with limited measurements [22]; a spatiotemporal graph model can also embed event-type, topology, and localization-cost knowledge in its loss [23].

Benchmark availability does not remove the contract problem. The pmuBAGE collection offers almost one thousand privacy-preserving generated event records learned from field data [24]; it is useful for standardized event analytics but does not by itself certify physical source holdout. Measurement integrity is also a distinct inference variable. Chatterjee *et al.* separate event-induced excursions from spurious outliers before robust PMU recovery [25], and Biswas *et al.* curate utility examples of missing, locked, quantized, and timing-affected synchrophasors [26]. Their evidence supports explicit integrity modeling instead of folding sensor failures into physical event classes.

These works show that limited PMU coverage is not by itself a new problem. The unresolved issue is the evaluation contract. Many classifiers learn a fixed list of asset labels, so testing another replica at a known source measures interpolation rather than source generalization. Zhai *et al.* reconstruct full state and localize line attacks from partial observations using an LSTM across multiple benchmark networks [7]. Rimkus *et al.* reconstruct missing trajectories without fault-location labels before applying a localization method [6]. These are direct conceptual competitors and prevent us from claiming to be the first to connect state reconstruction and localization. They do not, however, jointly represent the eight physical/integrity event states used here or test the exact held-source, diagnosability, and model-mismatch contract.

C. OBSERVABILITY, DIAGNOSABILITY, AND SENSOR VALUE

Functional observability asks whether a requested function of state can be inferred without full-state observability [27], [28]. PMU placement based on empirical observability Gramians similarly treats sensor value as task dependent

[29], [30]. For event sources, Jamei *et al.* derive placement-dependent performance limits for fault localization [31]; Conte *et al.* prove sparse-PMU fault-localization conditions that partition a radial grid into indistinguishable line clusters [32]; and Yildiz and Abur optimize sparse placement for event detection and identification [33]. These studies motivate the present separation between hidden-state reconstruction and pairwise hypothesis information. They also impose a restraint: one fixed eight-PMU set cannot support a general optimal-placement claim.

D. RESEARCH GAP AND CLAIM BOUNDARY

Table 1 exposes the remaining intersection rather than an absence of methods. The target is a source-agnostic physical hypothesis evaluated from sparse PMUs while (i) an unobserved state trajectory is estimated, (ii) physical interventions and measurement failures can coexist, (iii) source ambiguity is related to sensing geometry before a classifier is trusted, and (iv) generalization is tested on physical assets excluded from source-labeled training. A learned component is useful only if it improves model discrepancy or proposal efficiency without reverting to asset-label memorization.

The current evidence does not complete that intersection. It validates hidden-voltage estimation and physical recentering in PowerDynamics, and it audits a known-source event classifier plus source-transfer failure in ANDES. The proposed joint inference and diagnosability-to-posterior experiment remain the paper's next falsifiable step. This wording is intentionally narrower than the title's long-term objective and should be revised again if the end-to-end campaign is not completed before submission.

III. JOINT STATE, EVENT, AND SOURCE PROBLEM

A. HYBRID DAE AND ELECTRICAL MEASUREMENT MODEL

Let $x(t) \in \mathbb{R}^{n_x}$ denote differential states, $z(t) \in \mathbb{R}^{n_z}$ algebraic states, $u(t)$ scheduled inputs, $q(t)$ network configuration, and θ physical parameters. A physical intervention $a_h(t)$ is

TABLE 1. Capability comparison for the closest methodological families. “Held source” means that the physical asset is absent from source-labeled training or tuning. Entries describe published contracts, not a cross-paper performance ranking.

Work	State target	Event/source target	Physics mechanism	Sparse/partial sensing	Held source	Integrity separate
Katanic <i>et al.</i> [3]	Differential and algebraic DAE states	Unknown injectors, not event taxonomy	Incomplete nonlinear DAE and least squares	Yes	N/A	No
Ngo <i>et al.</i> [11]	Power-system states	None	Branch-current physics plus graph learning	Yes	N/A	No
Yue <i>et al.</i> [12]	Distribution-system state	Line outage	Grid-model-informed graph learning and robust SE	Yes	Not stated	No
Pandey <i>et al.</i> [4]	None	Detection, class, and source	Topology/equipment information	PMU network	Not stated	No
Yildiz and Abur [5]	None	Fault detection and location	OLS/CNN with strategic PMU placement	Yes	Not stated	No
Li <i>et al.</i> [23]	None	Multi-family disturbance source	Topology/type-constrained STGCN loss	Yes	Not stated	No
Rimkus <i>et al.</i> [6]	Missing PMU trajectories	Downstream fault localization	Graph reconstruction before diagnosis	Yes	No source labels for imputer	Corruption via imputation
Zhai <i>et al.</i> [7]	Full dynamical state	Line-attack location	Learned temporal dynamics	Yes	Across benchmark contracts	Attack target only
This paper	Hidden voltage field; state uncertainty	Eight event labels; physical and integrity sources	DAE hypotheses, projected information, residual learning proposed	Eight PMU buses	Required for final claim	Yes

indexed by hypothesis

$$h = (c, \ell, \tau, \gamma, d), \quad h \in \mathcal{H}, \quad (1)$$

where c is event family, ℓ is a bus or branch source, τ is onset, γ is severity, and d describes the allowed temporal profile. The plant is

$$\dot{x}(t) = f(x, z, u, q, a_h; \theta) + w_x(t), \quad (2)$$

$$\mathbf{0} = g(x, z, u, q, a_h; \theta), \quad (3)$$

$$y_k = M_k[h_{\mathcal{P}_k}(x_k, z_k, q_k; \theta) + b_k] + v_k. \quad (4)$$

Here M_k selects the channels that are available at time k , b_k is a measurement-corruption process, and v_k is measurement noise. A dropout changes M_k ; bad data change b_k ; neither is a physical input to (2)–(3). This separation permits a physical event and a measurement failure to coexist.

The algebraic network core is

$$I = Y_{\text{bus}}(q, \theta)V, \quad S_i = P_i + jQ_i = V_i I_i^*. \quad (5)$$

Writing $Y_{\text{bus}} = G + jB$ and $V_i = |V_i|e^{j\vartheta_i}$ gives

$$P_i = |V_i| \sum_j |V_j| [G_{ij} \cos \vartheta_{ij} + B_{ij} \sin \vartheta_{ij}], \quad (6)$$

$$Q_i = |V_i| \sum_j |V_j| [G_{ij} \sin \vartheta_{ij} - B_{ij} \cos \vartheta_{ij}], \quad (7)$$

where $\vartheta_{ij} = \vartheta_i - \vartheta_j$. A line outage changes the branch primitives and therefore Y_{bus} ; a bus fault adds a shunt-admittance intervention; generation and load changes modify the appropriate active/reactive injection equations. For a PMU at one terminal of branch (i, j) ,

$$I_{ij} = Y_{ff,ij}V_i + Y_{fi,ij}V_j, \quad I_{ji} = Y_{tf,ij}V_i + Y_{tt,ij}V_j. \quad (8)$$

The orientation, transformer tap, line charging, and shunt convention in (8) are part of the inference contract; replacing terminal current with bus-injection current changes the likelihood.

B. SPARSE PMUS AND REQUESTED STATE FUNCTION

Both evidence campaigns use PMU buses

$$\mathcal{P} = \{2, 5, 6, 10, 19, 22, 29, 39\}, \quad |\mathcal{P}| = 8. \quad (9)$$

The unobserved set is $\mathcal{U} = \{1, \dots, 39\} \setminus \mathcal{P}$, with $|\mathcal{U}| = 31$. The state-estimation campaign stacks rectangular voltage and controlled terminal-current components,

$$y_{\mathcal{P},k} = \text{col}_{p \in \mathcal{P}} [\Re V_{p,k}, \Im V_{p,k}, \Re I_{p\ell,k}, \Im I_{p\ell,k}], \quad (10)$$

and requests the hidden complex-voltage function

$$z_{\mathcal{U},k} = \text{col}_{i \in \mathcal{U}} [\Re V_{i,k}, \Im V_{i,k}] \in \mathbb{R}^{62}. \quad (11)$$

The event campaign also uses voltage angle, frequency, availability, and, in its longer-window implementation, current and ROCOF. Measurement semantics are kept campaign-specific rather than declaring channels equivalent across simulators.

C. EIGHT EVENT LABELS AS COMPOSABLE STATES

Table 2 defines the eight abnormal labels in the historical event bank. The scientific variables are not the label numbers. Classes 1–4 are single physical interventions; classes 5 and 7 are measurement-integrity states; classes 6 and 8 are compositions. Class 8 is a prescribed mixed event in the available simulations, not evidence of open-set recognition.

TABLE 2. Event ontology and historical ANDES inventory. Each source configuration has five replicas. “Physical source” and “integrity source” are separate outputs.

ID	Operational meaning	Physical intervention c	Physical source ℓ	Integrity state/source	Records
0	Normal operation	none	none	nominal	5
1	Bus fault	fault admittance	bus	nominal	195
2	Line outage	topology removal	branch	nominal	170
3	Generation change	active/reactive injection	generator bus	nominal	50
4	Load change	active/reactive withdrawal	load bus	nominal	90
5	Missing data	none	none	dropout / PMU	40
6	Generation change with missing data	generation change	generator bus	dropout / PMU	50
7	Bad data	none	none	corruption / PMU channel	40
8	Prescribed mixed disturbance	generation plus load changes	declared generator bus	measurement disturbance / PMU	50
Total					690

The internal decision at time k is

$$(\hat{z}_{\mathcal{U},k}, \Sigma_{z,k}, \hat{c}_k, \hat{\ell}_k^{\text{phys}}, \hat{m}_k, \hat{\ell}_k^{\text{meas}}, \mathcal{C}_k), \quad (12)$$

where m_k denotes the integrity state and $\mathcal{C}_k \subseteq \mathcal{H}$ is a credible candidate set. A location is empty when it does not apply. The candidate set is required because limited observability can make a forced Top-1 decision scientifically unjustified.

D. CORRECTNESS AND EVALUATION UNITS

State accuracy uses hidden-bus total vector error

$$\text{TVE}^{(r)} = \frac{1}{T_r |\mathcal{U}|} \sum_{k=1}^{T_r} \sum_{i \in \mathcal{U}} \frac{|\hat{V}_{i,k} - V_{i,k}|}{\max(|V_{i,k}|, \epsilon)}, \quad (13)$$

reported as a percentage. Event detection, event-family macro-F1, exact physical Top-1/Top-3, and integrity-source Top-1 use their own declared populations. The joint correctness indicator is one only when every applicable component of (12) is correct; a correct event family cannot conceal a wrong source. Confidence intervals use an independently generated trajectory, event parent, or rebuilt plant as the statistical unit, never individual 30-Hz samples.

The causal contract permits only $y_{0:k}$ at time k . A fixed-lag smoother, complete-window statistic, or centered temporal feature is retrospective and is labeled accordingly. Source generalization requires that physical source ℓ be absent from all source-labeled training and tuning, not merely that the test trajectory use a new random seed.

IV. FUNCTIONAL OBSERVABILITY OF THE HIDDEN VOLTAGE FIELD

A. LOCAL REDUCED MODEL

Linearizing the index-reduced DAE around a verified equilibrium and sampling at $\Delta t = 1/30$ s gives

$$\delta x_{k+1} = A_d \delta x_k + w_k, \quad (14)$$

$$\delta y_k = C \delta x_k + v_k, \quad (15)$$

$$\delta z_{\mathcal{U},k} = L_{\mathcal{U}} \delta x_k. \quad (16)$$

Here $\delta x_k \in \mathbb{R}^{114}$ contains the reduced differential coordinates, C maps them to the 32 voltage/current PMU channels, and $L_{\mathcal{U}}$ maps them to the 62 rectangular hidden-voltage components. These matrices are local: their validity can degrade

when the operating point or physical parameters move away from the linearization point.

For a horizon of N transitions, define

$$\mathcal{O}_N = \text{col}(C, CA_d, \dots, CA_d^{N-1}). \quad (17)$$

Full-state observability requires $\text{rank}(\mathcal{O}_N) = 114$. Hidden-voltage reconstruction requires a weaker, target-specific condition [27], [28].

Proposition 1 (Finite-horizon functional reconstruction): For the noise-free system (14)–(16), the initial target $L_{\mathcal{U}} \delta x_0$ is uniquely determined by the output sequence through horizon N if and only if

$$\ker \mathcal{O}_N \subseteq \ker L_{\mathcal{U}}. \quad (18)$$

Equivalently, there exists a matrix M_N such that $L_{\mathcal{U}} = M_N \mathcal{O}_N$.

Proof: Two initial states produce the same output history exactly when their difference belongs to $\ker \mathcal{O}_N$. Their target values agree for every such pair exactly when that difference also belongs to $\ker L_{\mathcal{U}}$, proving (18). In finite-dimensional spaces, $\ker \mathcal{O}_N \subseteq \ker L_{\mathcal{U}}$ is equivalent to the rows of $L_{\mathcal{U}}$ lying in the row space of $\mathcal{O}_N$, which yields $L_{\mathcal{U}} = M_N \mathcal{O}_N$. ■

Proposition 1 is a standard linear-algebraic characterization, not a novelty claim. Its role is to define the correct structural object for the tested output. It also makes clear that a low full-state rank does not by itself preclude accurate recovery of $z_{\mathcal{U}}$.

B. NUMERICAL FUNCTIONAL RESIDUAL

Exact rank statements are fragile under floating-point scaling, so the experiment uses the observable-subspace projector

$$\Pi_N = \mathcal{O}_N^\dagger \mathcal{O}_N \quad (19)$$

and the target residual

$$r_F(N) = \|L_{\mathcal{U}}(I - \Pi_N)\|_F. \quad (20)$$

For hidden bus i , the corresponding two-row map L_i gives

$$r_{F,i}(N) = \|L_i(I - \Pi_N)\|_F. \quad (21)$$

A smaller value means that fewer target directions project onto the numerically unobservable state subspace. It is a

structural diagnostic at the frozen equilibrium, not an empirical error or a probabilistic confidence interval.

The projector is obtained from the eigendecomposition of $\mathcal{O}_N^T \mathcal{O}_N$ using a fixed relative eigenvalue tolerance of 10^{-9} . Horizons $N \in \{0, 3, 10, 30, 60, 120, 180\}$ are evaluated. The $N = 180$ bus-level residuals and their ranking were exported before nonlinear reconstruction was scored. This preregistration permits a genuine theory-to-evidence test: the paper correlates $r_{F,i}(180)$ with per-bus TVE rather than defining the diagnostic after seeing the error pattern.

The same Gramian produces a trace-style noise amplification diagnostic under independent rectangular-voltage noise with standard deviation 10^{-3} pu. Because this bound depends on the local linearization and numerical rank tolerance, it is reported as structural context rather than as estimator performance.

C. EVENT-SOURCE DIAGNOSABILITY UNDER NUISANCE STATE

Functional observability concerns a state function; source localization concerns separation among interventions. Linearize candidate h over a horizon of H samples:

$$\delta x_{k+1} = A_k \delta x_k + B_{h,k} d_{h,k} + w_k, \quad (22)$$

$$\delta y_k = C_k \delta x_k + E_k b_k + v_k. \quad (23)$$

After stacking the available channels selected by M_k ,

$$Y_H = \mathcal{O}_H \delta x_0 + s_{h,H} + \mathcal{E}_H B_H + \varepsilon_H, \quad (24)$$

where $s_{h,H}$ is the candidate physical response, $\mathcal{O}_H$ is the time-varying observability matrix, and $\mathcal{E}_H B_H$ represents admissible integrity corruption. Let R_H be the stacked noise covariance. To avoid mistaking an uncertain initial condition for an event, define the R_H^{-1} -orthogonal nuisance projector

$$P_{\mathcal{O}}^\perp = I - \mathcal{O}_H (\mathcal{O}_H^T R_H^{-1} \mathcal{O}_H)^\dagger \mathcal{O}_H^T R_H^{-1}. \quad (25)$$

The usable signature is $\tilde{s}_{h,H} = P_{\mathcal{O}}^\perp s_{h,H}$. For candidates i and j with equal projected covariance, their local pairwise information is

$$D_{ij}^{(P)}(H) = \frac{1}{2} (\tilde{s}_{i,H} - \tilde{s}_{j,H})^T R_\perp^\dagger (\tilde{s}_{i,H} - \tilde{s}_{j,H}), \quad (26)$$

with $R_\perp = P_{\mathcal{O}}^\perp R_H (P_{\mathcal{O}}^\perp)^T$. This is the Gaussian mean-separation term after removing initial-state nuisance. It is not a new divergence.

Proposition 2 (Local indistinguishability): Assume the linearized model (24), equal prior probability, equal projected covariance, and no integrity pattern that differentiates candidates i and j . If $D_{ij}^{(P)}(H) = 0$, then the projected measurement distributions under i and j are identical. No decision rule based on the stated PMU history can attain an average error below $1/2$ for this pair.

Proof: Equation (26) is zero exactly when the two projected Gaussian means coincide on the support of $R_\perp$. Equal covariance then makes the distributions identical. With equal

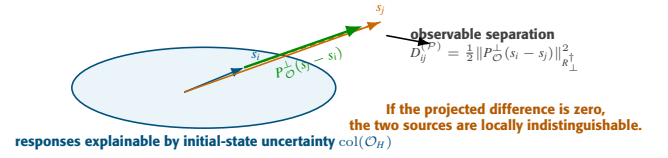


FIGURE 2. Local source-diagnosability geometry. Components within the initial-state response subspace are nuisance; only the projected difference separates two physical-source candidates. Coincident projected signatures imply the lower bound in Proposition 2.

priors, every measurable decision rule assigns the same observation law under both hypotheses, so its two conditional error probabilities sum to one. ■

Corollary 1 (Information cannot decrease when an independent PMU joins): If an added PMU block is conditionally independent of the existing measurements given each hypothesis and has a correctly specified covariance, then

$$D_{ij}^{(P \cup \{p\})}(H) = D_{ij}^{(P)}(H) + D_{ij}^{(p)}(H) \geq D_{ij}^{(P)}(H). \quad (27)$$

The corollary follows from additivity of Gaussian log-likelihood ratios. It gives a theoretical upper direction, not a guarantee that an arbitrary learned classifier improves when a PMU is added. Model error, dependence, or poor normalization can violate the assumptions.

For a source posterior $p(h \mid y_{0:k})$, define the smallest candidate set satisfying

$$\mathcal{C}_k(\alpha) = \arg \min_{\mathcal{C} \subseteq \mathcal{H}} \left\{ |\mathcal{C}| : \sum_{h \in \mathcal{C}} p(h \mid y_{0:k}) \geq 1 - \alpha \right\}. \quad (28)$$

A diagnosability-aware system abstains from exact Top-1 when $|\mathcal{C}_k| > 1$ or when the minimum information to a competing hypothesis falls below a development-set threshold. The pending joint experiment must test whether (26) predicts candidate-set size, posterior entropy, and source-resolution delay under PMU loss/join. The current event bank has not yet supplied that evidence.

V. PHYSICS-BASED STATE ESTIMATION AND JOINT INFERENCE

A. JOINT HYPOTHESIS-CONDITIONED OBJECTIVE

For candidate $h = (c, \ell, \tau, \gamma, d)$, let $X_{0:k}$ and $Z_{0:k}$ denote the differential and algebraic trajectory and let $B_{0:k}$ denote sparse measurement corruption. A physics-first fixed-horizon estimator can be written as

$$\begin{aligned} \mathcal{J}_h = & \|x_0 - \bar{x}_0\|_{P_0^{-1}}^2 + \sum_{t=0}^k \rho \left(\|r_{h,t}\|_{R_t^{-1}}^2 \right) \\ & + \sum_{t=0}^{k-1} \|x_{t+1} - F_h(x_t, z_t)\|_{Q^{-1}}^2 + \gamma_g \sum_{t=0}^k \|g_h(x_t, z_t)\|_2^2 \\ & + \lambda_b \sum_{t=0}^k \|b_t\|_1 - 2 \log p(h), \end{aligned} \quad (29)$$

where

$$r_{h,t} = M_t [y_t - h p_t(x_t, z_t) - b_t]. \quad (30)$$

The robust loss ρ limits the influence of unmodeled outliers, while the sparse term assigns persistent channel corruption to b_i instead of forcing the physical state to explain it. The event hypothesis changes the DAE through the mechanisms in Section III; thus event family and source are evaluated as physical counterfactuals rather than asset labels.

With a probabilistic model for every term, a Laplace or sequential approximation gives

$$p(h | y_{0:k}) \propto p(h) |H_h|^{-1/2} \exp[-\mathcal{J}_h(\hat{X}_h, \hat{Z}_h, \hat{B}_h)/2], \quad (31)$$

where H_h is an appropriate local Hessian on identifiable directions. If the determinant correction is omitted or the score is not calibrated, the output is an evidence score, not a posterior probability.

The learned component is deliberately restricted. The earlier engineered feature system can propose a small candidate set or learn a correction $\psi_\phi(r_{h,0:k})$ to the physical log evidence. It must be trained on candidate residuals rather than source identity alone and tested on held-out physical sources. This design distinguishes physics-guided signal features from a learned model discrepancy; only the latter modifies the inference objective. The required ablation is learned-only, physics-only, physics plus learned discrepancy, and the full diagnosability-aware decision rule.

B. NOMINAL ESTIMATOR HIERARCHY

The estimators form an evidence ladder rather than a collection of unrelated algorithms. B0 retains only the equilibrium, B1 adds instantaneous PMU conditioning, B2 adds causal state propagation, and B3 tests retrospective temporal information. When the equilibrium itself becomes stale, the separate E06-H level estimates a physically admissible operating point through nonlinear static MAP before any sequential extension is claimed. A full nonlinear Hybrid-DAE fixed-lag posterior is therefore an enclosing research formulation, not a result of this paper.

All nominal estimators use the same equilibrium y_0 , hidden-voltage center $z_{\mathcal{U},0}$, sampled dynamics A_d , PMU map C , and hidden-output map $L_{\mathcal{U}}$. This shared model isolates the value of measurement updating and temporal propagation.

The equilibrium baseline B0 returns

$$\hat{z}_{\mathcal{U},k}^{\text{B0}} = z_{\mathcal{U},0}. \quad (32)$$

It measures how much of the score can be obtained by remaining at the nominal power-flow solution.

The snapshot estimator B1 computes the minimum-variance linear update independently at every frame,

$$K_{\text{WLS}} = P_0 C^T (C P_0 C^T + R)^{-1}, \quad (33)$$

$$\hat{\delta x}_k^{\text{B1}} = K_{\text{WLS}}(y_k - y_0), \quad \hat{z}_{\mathcal{U},k}^{\text{B1}} = z_{\mathcal{U},0} + L_{\mathcal{U}} \hat{\delta x}_k^{\text{B1}}. \quad (34)$$

B1 uses network-constrained cross-covariance but no state carried from frame $k - 1$.

The causal Kalman estimator B2 applies the classical linear-Gaussian recursion [34]:

$$\hat{x}_{k|k-1} = A_d \hat{x}_{k-1|k-1}, \quad (35)$$

$$P_{k|k-1} = A_d P_{k-1|k-1} A_d^T + Q, \quad (36)$$

$$\nu_k = (y_k - y_0) - C \hat{x}_{k|k-1}, \quad (37)$$

$$S_k = C P_{k|k-1} C^T + R, \quad (38)$$

$$K_k = P_{k|k-1} C^T S_k^{-1}, \quad (39)$$

$$\hat{x}_{k|k} = \hat{x}_{k|k-1} + K_k \nu_k. \quad (40)$$

The covariance update uses the Joseph form. The hidden estimate is $z_{\mathcal{U},0} + L_{\mathcal{U}} \hat{x}_{k|k}$. B2 is causal because only $y_{0:k}$ enters the result at k .

B3 is an exact fixed-lag Gaussian smoother, following the linear Gaussian backward-recursion principle in [35], evaluated at lags $\ell \in \{0, 1, 3, 5, 10, 15, 30, 60\}$ frames. It uses measurements through $k + \ell$ to refine the estimate at k and is therefore retrospective by $\ell/30$ seconds. Dense and cached implementations agree numerically on representative lags. B3 tests whether temporal information beyond the causal update improves the hidden-voltage target; it is not used to support a real-time claim.

C. MARGINAL CALIBRATION AND COLORED DISCREPANCY

For the hidden two-component voltage error at bus i , B2 provides

$$\Sigma_{i,k} = L_i P_{k|k} L_i^T. \quad (41)$$

The original covariance is U_0 . A two-block temperature model U2 is fitted on an independent calibration split:

$$\Sigma_{i,k}^{\text{U2}} = T \Sigma_{i,k} T^T, \quad T = \text{diag}(\sqrt{\tau_{\Re}}, \sqrt{\tau_{\Im}}). \quad (42)$$

The temperatures minimize Gaussian negative log likelihood (NLL) on calibration data and are frozen before TEST. Coverage, NLL, normalized innovation squared (NIS), and temporal innovation correlation are reported separately; nominal marginal coverage alone is not called complete calibration.

The residual study considers a latent Gauss–Markov process

$$c_{k+1} = F_c c_k + \eta_k. \quad (43)$$

In D2-M, c_k enters the PMU measurement residual through a calibration-fitted low-rank basis. In D2-P, it enters a sparse physical process dictionary. Basis, rank, F_c , and latent covariance are selected using calibration records only. The experiment asks whether the observed low-rank structure improves reconstruction, hidden uncertainty, or innovation whiteness. It does not assume that a lower innovation autocorrelation is sufficient.

D. MODEL-ADEQUACY STATISTIC AND RECENTRING ABLATIONS

Let μ_ν and Σ_ν be the mean and shrunk covariance of B2 innovations on the nominal calibration set. The per-frame adequacy statistic is the five-frame trailing mean of

$$a_k = (\nu_k - \mu_\nu)^T \Sigma_\nu^{-1} (\nu_k - \mu_\nu). \quad (44)$$

Mismatch labels and hidden truth do not enter (44); hidden truth is used only after inference to label frames whose hidden TVE exceeds the frozen evaluation threshold. AUROC and AUPRC therefore measure whether observed innovations anticipate unacceptable hidden error, not whether the detector identifies a mismatch family.

Three variants diagnose the failure mechanism without merging their evidence contracts. R1 replaces the nominal equilibrium with the true plant equilibrium and is explicitly oracle. R2-A estimates an eight-dimensional grouped observed-bus voltage offset using a 60-frame trailing window, nominal matrices, ridge regularization, and only PMU measurements. It is a causal negative ablation: its center/state cross-covariance is neglected and it does not enforce the nonlinear AC equations.

The corrected physical static-MAP estimator addresses M6 operating-point mismatch at one equilibrium snapshot. Let

$$d = \text{col}(\Delta P_{PQ}, \Delta Q_{PQ}, \Delta P_{PV}) \in \mathbb{R}^{43} \quad (45)$$

collect active and reactive changes at 17 admissible PQ loads and active-power changes at nine PV buses. Voltage is parameterized as $V(\theta, \rho) = \exp(\rho + j\theta)$. Defining the scaled PMU residual as $e_y(V) = D_y^{-1}[h_P(V) - y_P]$, the estimator solves

$$\begin{aligned} (\hat{V}, \hat{d}) &= \arg \min_{V, d} \mathcal{J}(V, d), \\ \mathcal{J}(V, d) &= \|e_y(V)\|_2^2 + \gamma^2 \|g_{AC}(V, d)\|_2^2 + \frac{\lambda}{q_d} \|d\|_2^2. \end{aligned} \quad (46)$$

where h_P is the audited voltage/terminal-current operator, g_{AC} includes PQ balance, explicit PV-magnitude constraints, solved PV reactive power, and bus 31 as Slack, and D_y scales voltage and current components separately. The weights are $\gamma = 10^3$, $q_d = 1$, and $\lambda = 0.01$, selected on a disjoint DEV split; continuation initialization is then frozen for TEST. Neither the true nuisance vector nor the hidden voltages enter the TEST solve.

Because d may be non-unique, evaluation distinguishes nuisance information from target information. If J_{yd} and J_{zd} are the observed-PMU and hidden-voltage Jacobians with respect to d after eliminating voltage through the AC constraints, the local target residual is

$$r_{d \rightarrow z} = \frac{\|J_{zd} N(J_{yd})\|_F}{\max(\|J_{zd}\|_F, \epsilon)}, \quad (47)$$

where $N(J_{yd})$ spans the PMU-uninformed nuisance nullspace. Exact target identifiability would require a zero residual. The experiment instead tests the weaker empirical question: whether (46) reconstructs the hidden voltage accurately despite rank-deficient nuisance information, and whether its Laplace covariance is credible.

VI. VALIDATED BENCHMARK AND EXPERIMENTAL PROTOCOL

A. BENCHMARK PROVENANCE AND NUMERICAL GATES

The primary plant uses Julia 1.11.9, PowerDynamics.jl 5.0.0, and NetworkDynamics.jl 1.3.0 [36]. It contains 39 buses, 46

branches, 19 loads, 10 sixth-order Sauer–Pai machines, nine AVR Type-I controllers, and nine TGOV1 governors. The descriptor realization has 192 coordinates: 114 differential and 78 algebraic. These parameters are package-defined tutorial data. We therefore classify the plant as a numerically validated pedagogical benchmark rather than an authoritative representation of an external grid.

The validation ladder prevents estimator performance from hiding an upstream modeling error. A same-plant static translation was audited after correcting tap orientation, shunts, line charging, and full- Y_{bus} injection accounting. Final maximum discrepancies were 1.14×10^{-13} pu for a Y_{bus} element, 1.73×10^{-7} pu for voltage magnitude, 1.40×10^{-5} degrees for angle, and 1.71×10^{-3} MVA for terminal flow. The controlled PMU operator then achieved maximum complex-voltage and analytic/backend terminal-current discrepancies of 2.71×10^{-7} and 1.43×10^{-14} pu, respectively.

Dynamic validation checks the equilibrium, local modes, nonlinear trajectories, and linearization. The DAE equilibrium residual is 4.66×10^{-13} . After removing the reference mode, the least-damped physical pair is $-0.09818 \pm j0.42948 \text{ s}^{-1}$. Nonlinear Rodas5P simulations remain finite for the audited perturbations. Amplitude sweeps satisfy

$$\|y_{NL}(\epsilon) - y_{LIN}(\epsilon)\|_2 \propto \epsilon^p, \quad (48)$$

with fitted slopes 1.9820, 1.9987, and 2.0003 for representative state, governor-reference, and load perturbations, consistent with first-order linearization error.

B. NOMINAL RECONSTRUCTION AND CALIBRATION

The nominal dataset contains 20 development and 100 test trajectories, each sampled at 30 Hz for 3 s. Perturbations are normal-operation changes to governor references, loads, or their combination. The equilibrium, linear model, $P_0 = 10^{-2}I$, $Q = 10^{-6}I$, and $R = 10^{-6}I$ are frozen before TEST. B0, B1, B2, and all B3 lags are evaluated on the same trajectories. A separate 30-trajectory calibration split fits U1/U2 temperatures and low-rank discrepancy variants; TEST is not used for selection.

Hidden truth is used only for post-hoc metrics. The eight observed buses are excluded from every primary reconstruction score. The phasor prediction is formed as $V_0 + \Delta V$, and angle error is wrapped using $\text{atan2}(\sin \Delta\theta, \cos \Delta\theta)$. These rules correct an earlier exploratory implementation that incorrectly interpreted $\angle \Delta V$ as the absolute angle; no superseded result from that implementation is retained.

C. PHYSICAL MISMATCH CAMPAIGN

For each mismatch case, a fresh PowerDynamics plant is rebuilt from copied source tables, followed by power flow, dynamic initialization, and nonlinear simulation. The original package CSVs remain hash-identical. The seven families are network $R/X/B$ (M1), machine parameters (M2), governor parameters (M3), AVR parameters (M4), ZIP/load model (M5), operating-point redispatch (M6), and coupled mismatch (M7). Stress scale is

TABLE 3. Frozen experimental contracts. State and event campaigns retain separate simulators, splits, targets, and denominators.

Study	Frozen contract	Independent evaluation unit
Structural audit	Preregistered voltage/current target residual, horizons 0–180	31 hidden buses
Nominal reconstruction	20 DEV / 100 TEST; 91 frames at 30 Hz; B0–B3 frozen	Complete nonlinear trajectory
Nominal calibration	30 independent CAL trajectories; U1/U2 and discrepancy only	Complete nonlinear trajectory
Physical mismatch	Seven families, seven scales; B1/B2/U2 frozen	980 rebuilt plants
Breakpoint refinement	New seeds in transition bands; B2 frozen	175 rebuilt plants
Causal recentering	M1/M2/M6/M7; 160 DEV / 320 TEST; tune on DEV	Rebuilt plant trajectory
Physical static MAP	M6 PQ/PV/Slack basis; 30 DEV / 60 TEST; one snapshot	Rebuilt equilibrium
Known-source event diagnosis	Eight abnormal labels plus normal; 414 TRAIN / 138 DEV / 138 TEST	Complete ANDES trajectory
Source-transfer audit	Asset-disjoint folds; true family supplied; exact source	303 decisions from 101 event parents $\times$ 3 seeds

$m \in \{0, 0.25, 0.5, 0.75, 1.0, 1.25, 1.5\}$. These are engineering stress-test ranges, not probability priors.

All 980 main-grid cases and all 175 refinement cases passed power-flow, initialization, and time-domain acceptance. B1, B2, U0, U2, and the adequacy score are frozen from nominal data. Bootstrap intervals use seeds as independent units. Breakpoints in this paper are not inferred from the narrative report: the generated plots use the frozen per-case table directly because a prior refined breakpoint mixed aggregation definitions.

D. ORACLE, OFFSET, AND PHYSICAL-MAP ADAPTATION CONTRACTS

The oracle subset compares the frozen estimator with true-equilibrium recentering and true local relinearization on the same mutated trajectories. Oracle quantities are evaluation-only. The offset-only recentering study uses new DEV seeds 101–110 and TEST seeds 201–220, disjoint from all earlier datasets. Its window, cadence, regularization, and activation threshold are selected on DEV and frozen before TEST. No event labels, hidden state, or mismatch family enter the R2-A estimate.

The corrected physical static-MAP campaign is a separate M6-only contract. It uses 30 fresh DEV cases from seeds 601–610 and 60 TEST cases from seeds 611–630, with 20 TEST equilibria at each of $m \in \{0.5, 1.0, 1.5\}$. DEV selects the nuisance scale, regularization, and continuation initialization from a small fixed grid. TEST uses only the observed voltage and terminal-current phasors; hidden voltages are revealed afterward for scoring. A solution is accepted only when the nonlinear optimizer succeeds, all outputs are finite, and the maximum AC residual is at most 5×10^{-5} . Rejections remain in the denominator; the frozen rejection table is empty. A six-case subset is also solved with an independent constrained optimizer, and its partial convergence status is reported rather than upgraded to full solver equivalence.

All numerical tables and empirical figures are regenerated from the hashed snapshot under *generated/frozen*. The build script asserts 100 nominal test trajectories, 980 main-grid B2 cases, 60 corrected-MAP TEST cases with complete acceptance, rank-deficient nuisance information, and the preregistered adaptation-closure range before producing manuscript assets.

E. EVENT-DIAGNOSIS CAMPAIGN AND SIGNAL EVIDENCE

The separate event campaign uses ANDES [37] and the same eight PMU buses. It contains 138 source configurations: one normal condition, 39 fault buses, 34 outage lines, ten generation connections, 18 load buses, eight missing-data PMUs, ten generation-and-missing pairs, eight bad-data PMUs, and ten prescribed mixed-event configurations. Five replicas per configuration give 690 complete trajectories. Each physical trajectory lasts approximately 5 s, with the event scheduled at 2 s; missing or corrupted intervals last 0.7 s. Physical interventions are applied inside the simulator, whereas missing and corrupted values are applied to sampled measurements.

Three replicas per source configuration enter training, one development, and one test, producing 414/138/138 complete-trajectory splits. Consequently, the primary event benchmark evaluates new replicas at sources already represented in training. The separate asset-disjoint audit removes the physical source from all source-labeled fitting and tuning. Results from these contracts are reported side by side but never interpreted as the same generalization task.

The historical baseline is a hierarchy of extremely randomized trees. Its 136-feature version uses trailing voltage magnitude, angle, frequency, missingness, and a normalized disturbance score. The 448-feature version adds multi-scale statistics, adaptive residuals, and simultaneous contrasts among PMUs. A deterministic availability gate handles directly observed dropout. These are physics-guided features and typed decision heads; they are not a DAE-constrained learning architecture, and the ensemble’s leaf-vote scores are not treated as calibrated probabilities.

Figure 3 uses the first labeled episode of six representative families in the 89.65-min RAW0001 reference simulation. The curves are measured simulator outputs, not sketches or smoothed reconstructions. They show why no single channel suffices: a fault creates a short voltage sag, an outage redistributes current, injection changes have slower frequency/current responses, and measurement failures require availability or phase-consistency information.

VII. NOMINAL RECONSTRUCTION AND STRUCTURAL EVIDENCE

A. WHAT THE SPARSE PMUS RECOVER

Table 4 reports the frozen nominal TEST results. Remaining at the equilibrium (B0) gives 0.3402% mean hidden-bus TVE.

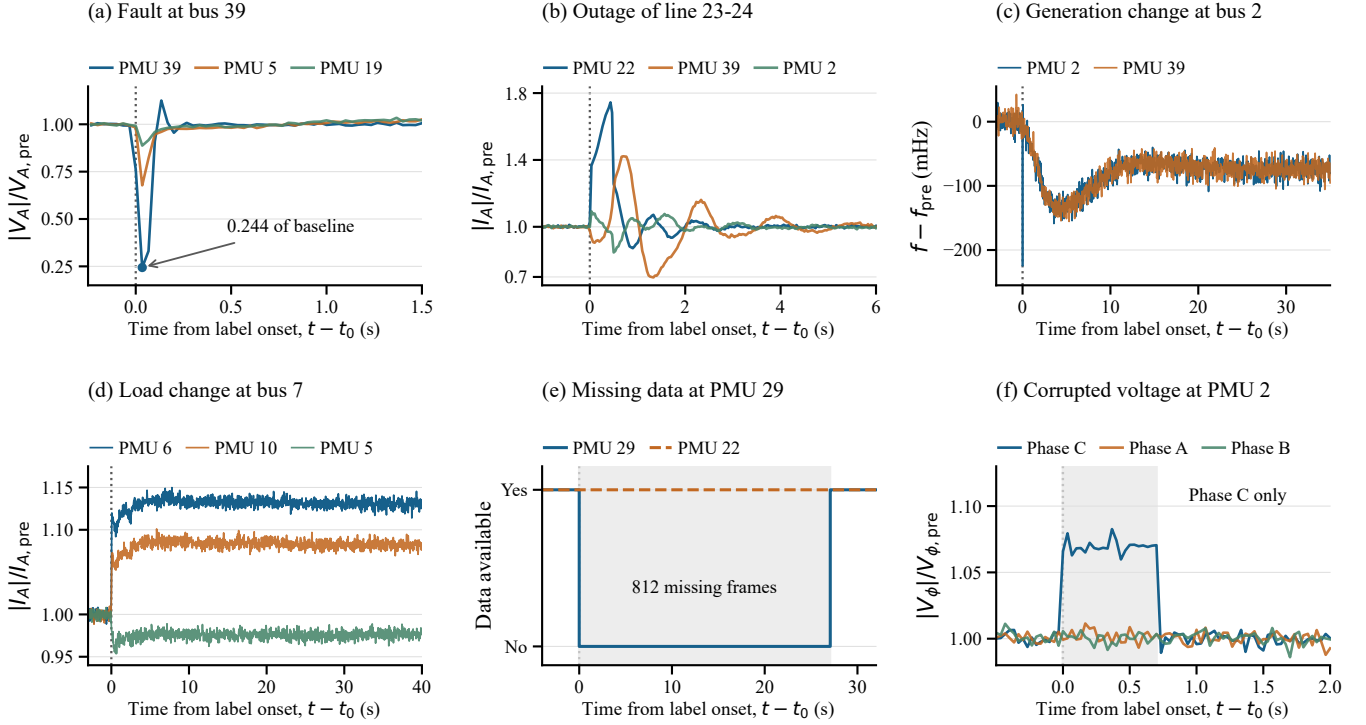


FIGURE 3. Representative measured event signatures from RAW0001. (a) Bus fault, (b) line outage, (c) generation change, (d) load change, (e) missing data, and (f) single-phase corruption. Dotted lines mark the first labeled sample; gray shading marks missing or corrupted intervals. Magnitudes are normalized by the preceding 3-s median, frequency is relative to its corresponding median, and time scales differ by panel. The plotted values and source-file hashes are frozen in the evidence snapshot.

Applying the same PMU evidence through the snapshot electrical update (B1) reduces this to 0.009904%, showing that the dominant nominal gain comes from the measurement-constrained physical map rather than temporal recurrence. The causal Kalman recursion (B2) reaches 0.009634%. Because both estimators score the same 100 trajectories, the relevant comparison is the paired difference: B2 minus B1 is -0.000270 percentage points (95% paired-bootstrap CI, -0.000359 to -0.000179) and B2 wins on 69.0% of trajectories. The temporal contribution is therefore consistent but small in absolute magnitude, a 2.7% reduction in the mean.

Figure 4(a) makes the trajectory distributions and scale separation visible, while panel (c) shows the paired differences rather than comparing overlapping marginal intervals. The equilibrium baseline is more than one order of magnitude worse than either measurement-updated estimator. This is not a comparison against learned or nonlinear DAE estimators; it is a controlled decomposition of equilibrium centering, instantaneous PMU conditioning, and causal temporal propagation within one local model.

B. DOES FUNCTIONAL STRUCTURE PREDICT ERROR?

At horizon 180, the finite-horizon matrix has numerical rank 39 of 114 and the aggregate target residual is nonzero. Accordingly, the experiment does not declare exact full-state or exact functional observability. The preregistered bus residual nevertheless correlates with B2 TVE at $\rho = 0.5625$ (Spear-

man), as shown in Fig. 4(b). A trajectory-independent, bus-level bootstrap gives an exploratory 95% interval of [0.271, 0.751], the two-sided permutation p -value is 0.0012, and Kendall $\tau = 0.376$. The same qualitative relation appears for B1. These secondary statistics support the narrower claim that the local structural diagnostic ranks empirical difficulty better than chance under the nominal benchmark.

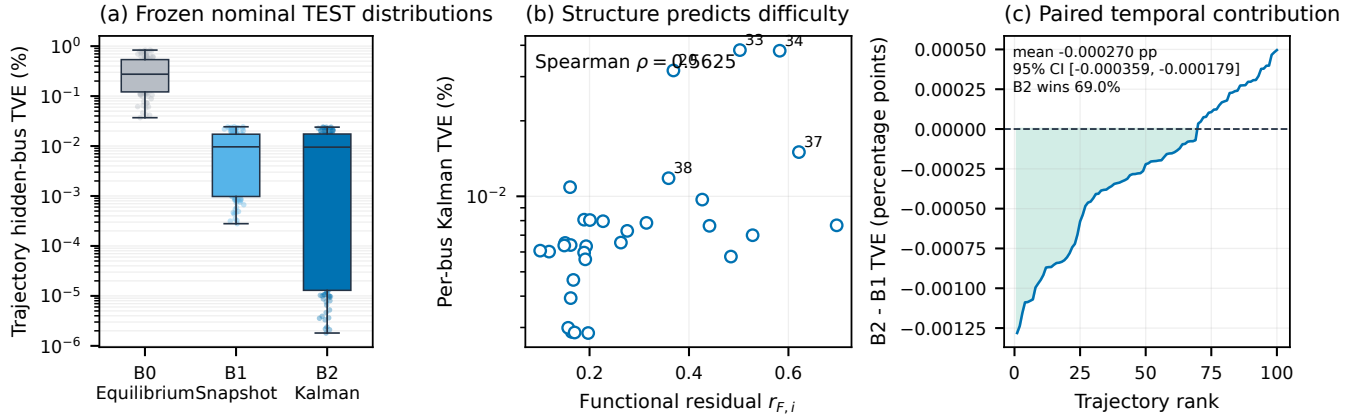
The relationship is informative but incomplete. Bus 36 has the largest structural residual, whereas the largest B2 errors occur at buses 33, 34, 20, 37, and 38. Thus the structural projector captures one source of difficulty but not the nonlinear excitation distribution, estimator covariance, or all numerical conditioning effects. Electrical distance to the nearest PMU has not yet been computed from the same audited Power-Dynamics network, so the present analysis does not establish that $r_{F,i}$ adds information beyond that confounder. This counterexample and unresolved control are why the residual is reported as a diagnostic rather than converted into a binary observability certificate.

C. DOES RETROSPECTIVE SMOOTHING HELP?

Across lags from one frame to 2 s, the exact fixed-lag smoother never improves mean TVE over lag zero; every positive lag is slightly worse in the frozen results. Figure 5(a) shows the full curve. In this regime, current voltage and terminal-current phasors already dominate the hidden-voltage update, and additional future samples do not compensate for

TABLE 4. Nominal reconstruction on 100 frozen nonlinear TEST trajectories. Metrics use the 31 unobserved buses only; TVE is reported in percent and intervals bootstrap independent trajectories.

Method	Complex RMSE	$ V $ RMSE	Angle RMSE ($^\circ$)	Mean TVE (%)	95% TVE interval (%)
B0 equilibrium	3.486e-03	2.496e-05	0.19497	0.340200	[0.293231, 0.387428]
B1 snapshot WLS	1.529e-04	5.091e-05	0.00821	0.009904	[0.008342, 0.011514]
B2 causal Kalman	1.338e-04	4.235e-05	0.00722	0.009634	[0.008006, 0.011302]

**FIGURE 4.** Nominal reconstruction and structural evidence. (a) Trajectory-level hidden-bus TVE distributions on the same 100 nonlinear TEST trajectories. (b) Per-bus causal Kalman TVE versus the horizon-180 functional residual exported before TEST scoring; labels identify the five buses with largest empirical TVE. (c) Sorted paired B2-minus-B1 TVE differences; negative values favor the causal estimator, and the annotation reports the paired bootstrap interval and win rate.

the local-model approximation. The negative result rules out latency as a useful trade for accuracy under the tested nominal perturbations.

VIII. UNCERTAINTY, PHYSICAL MISMATCH, AND ADAPTATION

A. MARGINAL CALIBRATION DOES NOT WHITEN INNOVATIONS

The nominal B2 covariance is highly conservative: U0 covers essentially all real and imaginary errors at the nominal 95% level. Fitting two temperatures on 30 independent CAL trajectories improves TEST hidden NLL from -13.477 to -16.184 and gives 93.9% and 94.2% marginal 95% coverage for real and imaginary components, respectively. This is useful marginal calibration, but it leaves the innovation lag-one autocorrelation unchanged at 0.602 and the Ljung–Box test rejects whiteness. Figure 5(b)–(c) and Table 5 therefore separate interval coverage from temporal model adequacy.

The calibration residual is strongly low rank: its first two principal components explain 95.7% of variance and four explain 99.2%. That empirical fact does not guarantee a useful generative correction. The rank-four measurement Gauss–Markov model reduces lag-one autocorrelation from 0.602 to 0.540, but worsens TVE, produces hidden NLL 14.991, and undercovers both components. The rank-one process placement leaves both TVE and autocorrelation effectively unchanged. These ablations reject the claim that low-rank colored PMU residuals, inserted without stronger physical structure, improve hidden-state uncertainty.

B. WHICH PHYSICAL MISMATCHES MATTER?

The 980-case rebuilt-plant campaign reveals sharp family dependence (Fig. 6 and Table 6). At $m = 1.5$, network mismatch reaches $16.99\times$ the pooled nominal median and operating-point mismatch reaches $8.32\times$. The coupled family is smaller at $1.602\times$, while machine, governor, AVR, and load-model families remain near the nominal median for the tested perturbations. A blanket claim of robustness would therefore be misleading: the estimator is insensitive to several tested component shifts but fails materially when the algebraic network map or equilibrium center changes.

Nominal covariance repair does not transfer automatically. Averaged across all mismatch cases, U0 has 92.9% 95% coverage and nominally fitted U2 has only 47.5%. The reduction reflects physical shift, not a calibration/test leak: U2 remains frozen from nominal CAL data. This result limits the interpretation of Section V: nominal dispersion scaling is not robustness to a changed plant.

C. ADEQUACY SIGNALS MISMATCH AND PHYSICAL STATIC MAP RECOVERS M6

The observed-PMU innovation statistic detects many cases in which hidden TVE is unacceptable. Macro AUROC/AUPRC are 0.901/0.779 at $m = 0.5$ and 0.928/0.787 at $m = 1.5$ [Fig. 6(c)]. Undefined family-level AUROCs, caused when every frame has the same frozen adequacy label, are omitted from the macro mean; AUPRC remains reported. Consequently, the statistic supports model-adequacy triage, not mismatch-family classification or guaranteed alarm calibration.

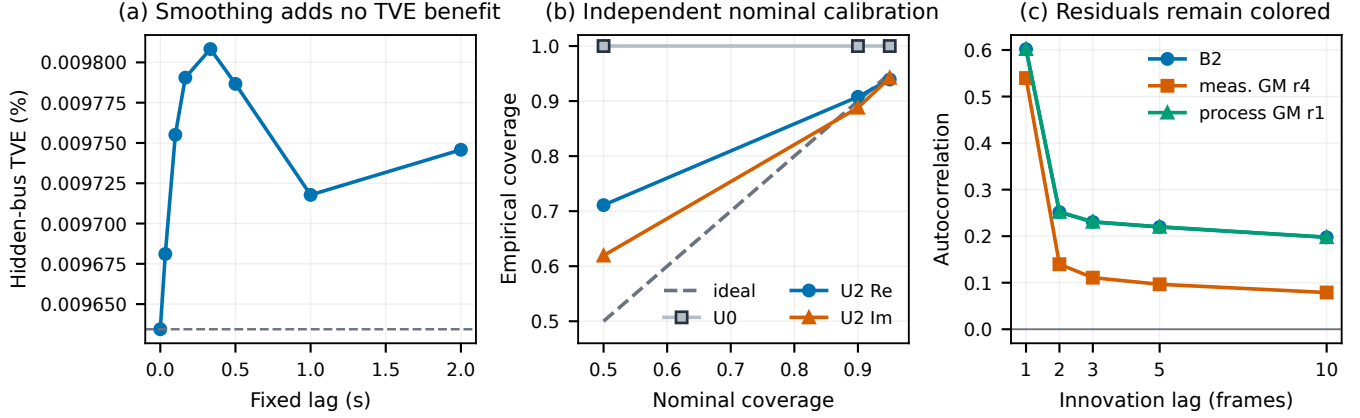


FIGURE 5. Temporal and uncertainty diagnostics on the frozen nominal TEST set. (a) Exact fixed-lag smoothing; lag zero is the causal Kalman result. (b) Marginal real-component coverage for U0 and real/imaginary U2 blocks against nominal coverage. (c) innovation autocorrelation for the original estimator and calibration-selected low-rank discrepancy placements.

TABLE 5. Frozen TEST uncertainty and colored-discrepancy ablations. U2 is fitted on a disjoint 30-trajectory calibration set; Gauss–Markov (GM) ranks are selected on that calibration set.

Variant	Hidden NLL	95% cov. Re (%)	95% cov. Im (%)	Innovation ACF(1)	TVE (%)
U0 original	-13.477	100.0	100.0	0.602	0.009634
U2 Re/Im temperature	-16.184	93.9	94.2	0.602	0.009634
D2-M measurement GM r4	14.991	66.6	48.1	0.540	0.009788
D2-P process GM r1	-13.475	100.0	100.0	0.602	0.009638

TABLE 6. Median B2 hidden-bus TVE at the largest mismatch scale, $m = 1.5$. Ratios use the 140-case main-grid nominal median of 0.014744%.

Mismatch family	TVE (%)	Ratio
M1 network	0.250430	16.985×
M2 machine	0.014744	1.000×
M3 governor	0.014743	1.000×
M4 AVR	0.014744	1.000×
M5 load model	0.014761	1.001×
M6 operating point	0.122641	8.318×
M7 coupled	0.023615	1.602×

The earlier R2-A offset-only ablation remains scientifically useful because it isolates what does not solve the problem. Across nonnominal identifiable M1, M6, and M7 cases it closes only 0.4% of the oracle gap, despite passing its nominal-safety gate. That failure does not imply that adaptation from observed PMUs is impossible; it shows that grouped voltage offsets without AC balance and a rebuilt physical center are inadequate.

The corrected static-MAP experiment directly tests that distinction for M6. All 60 TEST solves satisfy the frozen convergence gate. From $m = 0.5$ to 1.5, the nominal-center median TVE rises from 0.19885% to 0.57795%, whereas the physical MAP gives 0.00525% to 0.00974% [Fig. 7(a)]. Paired oracle-gap closure is 97.4%, 98.1%, and 98.3% across the three scales, with bootstrap intervals reported in Table 7. Median solve time is 61.7 ms and the 95th percentile is 76.0 ms. These timings characterize the implemented static solve only; they do not establish end-to-end real-time opera-

tion.

The nuisance information matrix has median rank 25 for 43 coordinates, leaving an 18-dimensional nullspace. Nevertheless, the requested hidden-voltage field is recovered with two orders of magnitude less error than the frozen center. This is empirical evidence for target recovery despite nuisance non-uniqueness, not a proof of exact functional identifiability: the local hidden-voltage residual is nonzero at 0.0419. The distinction links the adaptation result to Section IV without claiming that the injection vector itself is recovered.

Point accuracy also does not certify posterior uncertainty. The nullspace-aware Laplace covariance attains 100% marginal 95% coverage, yet its median NEES-like statistic increases from 111.6 to 294.2 [Fig. 7(c)]. The independent constrained solver also converges on only part of its six-case subset. The E06-H conclusion is therefore strong for corrected M6 point reconstruction, partial for local target structure and uncertainty, and not established for M1 network mismatch, M7 coupled mismatch, or sequential online adaptation.

D. HOW FAR DOES THE KNOWN-SOURCE EVENT BASELINE GO?

Table 8 reports the audited sequential event baseline. On the known-source TEST split, the 448-feature hierarchy with the availability gate attains detection F1 0.989, event macro-F1 0.905, physical Top-1 0.592, physical Top-3 0.719, and integrity Top-1 0.974. The expanded representation improves physical Top-1 over the 136-feature hierarchy by 9.6 percentage points (95% paired interval, 6.9–12.5). This isolates the

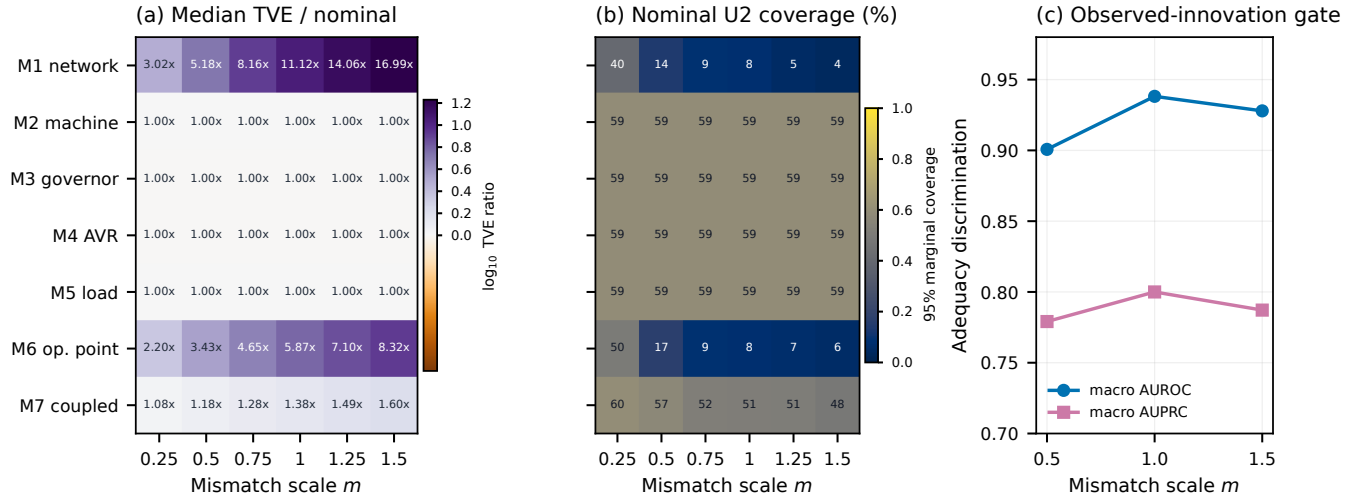


FIGURE 6. Physical-mismatch atlas for the frozen estimator. (a) Median B2 hidden-bus TVE relative to the pooled $m = 0$ median for every family-scale cell; annotations show the untransformed ratio and color encodes its base-10 logarithm. (b) Mean nominal-U2 marginal 95% coverage under the same rebuilt plants. (c) Macro adequacy discrimination from observed innovations; undefined family-level AUROCs are omitted from the macro mean. Each family-scale cell contains 20 independently rebuilt plants.

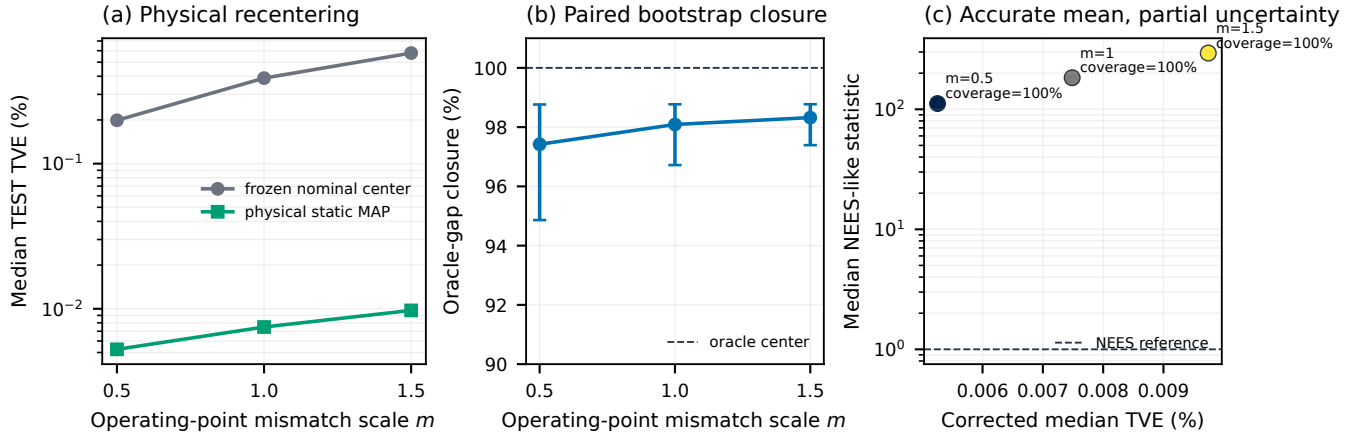


FIGURE 7. Corrected physical static recentering for held-out M6 operating-point mismatch. (a) Median hidden-bus TVE for the frozen nominal center and the physical static MAP. (b) Paired oracle-gap closure with 95% bootstrap intervals over 20 independent TEST cases per scale; the oracle-center reference is 100%. (c) Corrected point error versus the median joint NEES-like diagnostic. Marginal 95% coverage is 100% at each scale, but the joint statistic remains far above its nominal reference.

value of multiscale dynamics, adaptive residuals, and PMU contrasts under known-source training; the availability rule mainly improves integrity handling.

Figure 8(a) separates detection, family classification, physical localization, and integrity localization. The gap between detection F1 and physical Top-1 confirms that a recognizable disturbance need not have a uniquely recoverable source. False alarms decrease to 1.45 episodes per normal-labeled minute, but the normal exposure is only about 7.15 min per split. That rate is evidence of alarm burden in a short synthetic stream, not an operational false-alarm guarantee.

E. DOES LOCALIZATION TRANSFER TO UNSEEN PHYSICAL SOURCES?

No. With the physical source excluded from source-labeled fitting and the true event family supplied, only 3 of 303

decisions across 101 held-source cases and three model seeds are exact, a rate of 1.0%. This is not a detector failure or a confusion among event families: it is direct evidence that the learned location heads and candidate descriptors retain source-specific associations. The negative result invalidates any zero-shot-source claim for the legacy system.

Transfer to RAW0001 is also weak. The basic candidate model classifies an average of 2.7 of 12 event episodes and localizes 0.3 of five physical episodes across three seeds. The simulator bank omits some channels used by the longer-window implementation, so the transfer deficit cannot be assigned to one feature family alone. It nevertheless shows that improving the internal simulation score does not establish cross-record reliability.

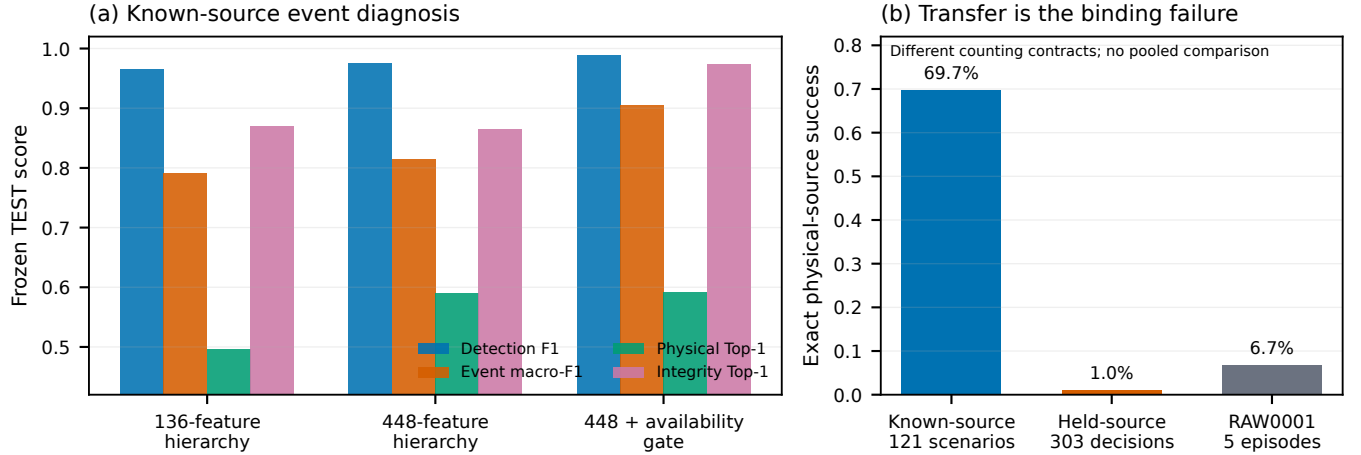
The event evidence therefore answers a precise baseline question: supervised, physics-guided feature engineering is

TABLE 7. Corrected physical static recentering on 60 held-out M6 operating-point cases. Gap-closure intervals bootstrap the 20 independent TEST cases at each scale. Rank and nullity refer to the 43-dimensional nuisance basis; runtime is per solve.

m	n	Frozen TVE (%)	Corrected TVE (%)	Gap closure [95% CI] (%)	Rank	Nullity	Median NEES	Runtime (ms)
0.5	20	0.19885	0.00525	97.4 [94.9, 98.8]	25/43	18	111.6	57.6
1.0	20	0.38834	0.00749	98.1 [96.7, 98.8]	25/43	18	183.3	61.1
1.5	20	0.57795	0.00974	98.3 [97.4, 98.8]	25/43	18	294.2	69.2

TABLE 8. Audited sequential event-diagnosis baseline on the known-source TEST split, mean over three fitted seeds. The independent split unit is the complete trajectory; timestamp-level scores retain the denominators of the frozen evaluator.

Configuration	Detection F1	Event macro-F1	Physical Top-1	Physical Top-3	Integrity Top-1	False alarms/min
136-feature hierarchy	0.965	0.791	0.496	0.646	0.870	4.01
448-feature hierarchy	0.975	0.815	0.590	0.717	0.865	1.45
448 features + availability	0.989	0.905	0.592	0.719	0.974	1.45

**FIGURE 8.** Event evidence and its transfer boundary. (a) Frozen known-source TEST metrics for the compact hierarchy, expanded hierarchy, and availability-gated model. (b) Exact physical-source success under three distinct counting contracts: known-source completed scenarios, source-held-out decisions, and RAW0001 physical episodes. The bars in (b) are deliberately not pooled; they expose the change in task difficulty and data semantics.

effective for detection and known-source diagnosis, but it does not solve source-generalizable physical inference. The result motivates the candidate-conditioned residual objective in (29) and the projected information test in (26); it is not itself evidence that those proposed components work.

IX. DISCUSSION AND LIMITATIONS

The two campaigns establish different parts of the joint problem. The PowerDynamics experiments show that eight voltage/current PMUs can reconstruct the requested 31-bus voltage field accurately under the tested nominal perturbations even though the 114-state local model is far from full rank. The ANDES experiments show that the same sensor buses support strong abnormal-event detection and moderate known-source localization across eight event labels. Neither result implies the other: low state TVE does not guarantee source separation, and high detection F1 does not imply a correct hidden state or physical source.

Functional observability and source diagnosability explain the distinction. Proposition 1 removes state directions that do not affect the requested voltage function. Proposition 2

instead asks whether two interventions remain different after uncertain initial state is projected out. These are local linear results. They do not certify a nonlinear DAE over large faults, changing topology, or unmodeled controllers. Their scientific value depends on a prospective empirical bridge: the functional residual should predict reconstruction error, and pairwise information should predict candidate-set size, posterior entropy, or time to source resolution.

The first bridge has partial evidence. The preregistered functional residual correlates with bus-level error, but the exceptions show that excitation, numerical tolerance, estimator covariance, and electrical distance still matter. The second bridge remains untested. The source-held-out collapse is consistent with poor physical separation or source memorization, but the present artifacts cannot distinguish those explanations. Computing (26) before training and comparing it with held-source errors is therefore more informative than adding another classifier layer.

The state-adaptation result narrows the role of identifiability. On M6 operating-point shift, the physical static MAP closes 98.1% of the oracle-center gap although only 25 of

43 nuisance directions are locally informed. Accurate recovery of a target function can coexist with non-unique nuisance coordinates. This does not extend automatically to line outages, faults, or sequential event-conditioned estimation. The successful solver uses one equilibrium snapshot, and its Laplace covariance remains severely under-dispersed in the joint NEES-like diagnostic.

The legacy event baseline identifies both reusable mechanisms and a hard failure. Multiscale temporal summaries, adaptive residuals, and simultaneous PMU contrasts improve known-source physical ranking by 9.6 percentage points. Availability should remain an explicit integrity input because a missing measurement is observed directly. By contrast, source identities and endpoint descriptors permit memorization: with complete physical-source holdout, exact localization falls to 1.0%. A stronger hybrid should therefore compute engineered features on candidate physical residuals, $y - \hat{y}_h$, and compare them with a physics-only likelihood. This reuses domain knowledge without treating training assets as the candidate universe.

Five limitations block a submission-ready joint-method claim. First, the state and event evidence use different simulators and are not an end-to-end common-contract comparison. Second, the PowerDynamics plant and RAW0001 are simulated; no field or hardware-in-the-loop validation is included. Third, the event bank has short, fixed-onset trajectories and about 7.15 min of normal-labeled exposure per split, which is insufficient for operational false-alarm claims. Fourth, PMU loss/join has not been tied prospectively to the information measure, and topology/source transfer remains unvalidated. Fifth, the proposed posterior in (31) has not been calibrated; classifier votes and uncorrected evidence scores must not be called probabilities.

The next frozen campaign should use one audited plant and measurement operator across all four baselines: learned-only, physics-only, physics plus learned residual discrepancy, and diagnosability-aware inference with abstention. Complete event parents and physical sources must remain disjoint across training, development, calibration, and test. The protocol should vary operating point, parameter mismatch, PMU loss/join, and at least one simulator; expose a long normal stream; report false alarms per hour, event recall, joint event-plus-source correctness, candidate-set coverage and size, source-resolution delay, TVE, NLL/ECE, runtime, and failed solves. Until that campaign is complete, this manuscript is a consolidated scientific draft with validated components and an explicit falsifiable gap, not a completed demonstration of joint superiority.

X. CONCLUSION

This paper formulated sparse-PMU monitoring as joint inference of a hidden voltage field, physical event, physical source, and measurement-integrity state. Functional observability characterizes recoverability of the requested state function; projected pairwise information gives a local limit on source separation after initial-state nuisance is removed.

The consolidated evidence identifies both capability and boundary. On 100 frozen PowerDynamics trajectories, the causal estimator attains 0.009634% mean hidden-bus TVE, and physical static MAP closes 97.4%–98.3% of the operating-point error gap on 60 held-out plants despite nuisance rank 25/43. On 690 separate ANDES trajectories, the known-source event baseline reaches detection F1 0.989, event macro-F1 0.905, and physical Top-1 0.592. Yet complete source holdout yields only 3/303 exact decisions, and transfer to RAW0001 is weak.

The central consequence is therefore not a claim of finished joint superiority. Sparse PMUs can support accurate target-state reconstruction and strong known-source detection, but source-generalizable diagnosis requires physical candidate responses, an explicit integrity model, and diagnosability-aware abstention. A common-contract end-to-end campaign must establish that result before the title's full joint-inference claim is suitable for submission.

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